Parth Khandagale

Chemical Engineer | Materials Researcher

Final-Year PhD Candidate, University at Buffalo | Buffalo, New York

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RESEARCH PROFILE

Chemical engineer and materials researcher developing multifunctional nanomaterials for electromagnetic shielding, neutron attenuation, and rare-earth photonics. Research connects materials synthesis, advanced characterization, and application-led testing, with a particular focus on thin functional systems in which composition and structure enable electromagnetic, radiation-shielding, and optical performance.

EDUCATION

University at Buffalo, SUNY — **Aug 2023–Present**
PhD in Chemical Engineering | Swihart Research Group — *Buffalo, New York*
GPA: 3.79/4.00

Institute of Chemical Technology — **Aug 2018–Jul 2023**
Integrated MTech in Chemical Engineering | Minor in Energy Engineering — *India*
GPA: 9.32/10.00

SELECTED RESEARCH PROJECTS

Ultrathin multifunctional shielding — 2025 | Manuscript under review

MXene–exfoliated boron binder-free hybrid films for simultaneous EMI and thermal-neutron shielding. Reported performance: 40 dB EMI shielding and 3% thermal-neutron shielding in 3 μm films.

Dual-mode anti-counterfeiting — Jan–Dec 2025 | Manuscript in preparation

Optically up-converting and electrically conductive RFID security tags with lithographic patterning for microscale features in nearly transparent optical and electronic films.

Rare-earth MXene chemistry — Jan–Dec 2024

Etching-strategy development for $(Mo_{2/3}Nd_{1/3})_2CT_x$, connecting selective chemical processing with rare-earth-containing two-dimensional materials.

Photocatalytic wastewater treatment — Sep 2022–Aug 2023

Graphitic carbon nitride photocatalysts for visible-light methylene-blue degradation, with a reported degradation efficiency of 66.67% under the thesis experiment's conditions.

RESEARCH EXPERIENCE

University at Buffalo — **Aug 2023–Present**
PhD Researcher, Swihart Research Group — *Buffalo, New York*

- Develop MXene-based nanocomposites for lightweight electromagnetic-interference and neutron-radiation shielding.
- Designed binder-free MXene–exfoliated boron hybrid films and conducted application-led performance analysis.
- Explore etching strategies for neodymium-containing MXenes derived from lanthanide-containing in-plane MAX phases.
- Develop dual-mode anti-counterfeiting concepts combining up-converting films, electrical conductivity, RFID tags, and microscale lithographic patterning.
- Apply diffraction, spectroscopy, microscopy, thermal analysis, surface-area analysis, and physical-property measurements.

Institute of Chemical Technology — **Sep 2022–Aug 2023**
Graduate Researcher | Advisor: Dr. Parag R. Nemade — *Mumbai, India*

- Synthesized graphitic carbon nitride photocatalysts for visible-light degradation of methylene blue.
- Connected materials synthesis and optical response with environmental-engineering applications and reviewed industrial treatment approaches for landfill leachate.

TECHNICAL EXPERTISE

Structure & Chemistry	XRD, XPS, Raman spectroscopy, FTIR, and XRF
Imaging	SEM–EDS, TEM, and AFM
Physical Properties	TGA, DSC, BET surface-area analysis, PPMS, UV–Vis spectroscopy, and differential centrifugal sedimentation (DCS)
Synthesis & Fabrication	MXene synthesis; etching-strategy development; rare-earth materials chemistry; binder-free hybrid films; lithographic patterning; graphitic carbon nitride synthesis; specialty-chemical synthesis
Engineering & Analysis	Gas-inducing impeller design and fabrication; mixing-time and mass-transfer analysis; photocatalytic treatment; process cost modeling; Monte Carlo neutron simulations; electrochemical testing; data interpretation
Software	MATLAB, Python, and Microsoft tools

SELECTED PUBLICATIONS

2026. P. Khandagale, S. K. Arachchige, C.-T. Chen, et al. "Comparative Study of Five Distinct Etching Strategies to Synthesize 2D $(Mo_{2/3}Nd_{1/3})_2CT_x$ i-MXene From $(Mo_{2/3}Nd_{1/3})_2AlC$ i-MAX Phase.". *Small*. DOI: 10.1002/smll.76060

2026. "Rare-Earth-Modified Luminescent Two-Dimensional Nanosheets Derived from Oxidized Mo_2CT_x MXene.". *2D Materials*. Journal article

2026. "Linear and Nonlinear Optical Response of Rare-Earth i-MAX Carbides.". *ACS Applied Optical Materials*. Journal article

2023. "Advanced Oxidation-Based Pretreatment of Wastewater for Enhanced Biofuel Production.". *Biomass and Biofuels, CRC Press*. Book chapter

2021. "Overview and Advances in Wastewater Treatment Plant with Focus on Pharmaceutical Industry Wastewaters.". *Indian Chemical Council*. Article

INDUSTRY AND RESEARCH EXPERIENCE

Vesuvius–Foseco India Ltd. **Jul–Oct 2022**
Project Intern *Pune, India*

- Developed process cost models to optimize raw-material purchasing and centralized a global supplier database for imported raw commodities.

Amogh Chemicals **2022**
Research & Synthesis Intern *Badlapur, India*

- Supported specialty-chemical synthesis and process calculations; prepared a nitro-substituted benzoic acid intermediate with a reported yield of 94.14%.

Snowtech Equipment Pvt. Ltd. **Sep–Dec 2021**
Project Intern *Mumbai, India*

- Designed and fabricated gas-inducing impellers; estimated critical impeller speed and evaluated mixing time and gas–liquid mass transfer.

Institute of Chemical Technology **2020**
Research Intern | Prof. Parag Gogate *Mumbai, India*

- Studied pharmaceutical-wastewater treatment using advanced oxidation and electrochemical methods and examined effluent-treatment-plant economics.

Godrej Industries Ltd. **Sep–Dec 2019**
Student Intern *Valia, India*

- Analyzed glycerine and fatty-acid production by fat splitting, hydrogen production by steam reforming, and sodium lauryl sulphate production.

Approcopp Engineering **2019**
Internship Trainee *Jalna, India*

- Studied refrigeration systems, thermodynamic cycles, welding processes, and copper-capillary design for refrigerant transport.

TEACHING AND MENTORING

- Served as a teaching assistant for two semesters across Material Science and Engineering, Applied Chemical Kinetics, and Chemical Engineering Mathematics and Computation, supporting 95 combined course enrollments through problem-solving sessions, concept clarification, and grading.
- Mentored an undergraduate project on M_4AlX_3 MAX phases and corresponding $M_4X_3T_x$ MXenes.
- Mentored a graduate project on MXene-based battery electrodes; both students completed oral presentations, and the graduate project culminated in an MS thesis.